

Machine Learning Applications for Health

健康领域的机器学习应用

COMP90089 (2026) - Lecture 3 - Data Sources

COMP90089 (2026) - 第 3 讲 - 数据源

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Today's Agenda: Data Sources

今日议程：数据源

- Sources of clinical data 临床数据来源
- Some openly (and not so openly) available datasets

一些公开（以及不太公开）可获取的数据集

- MIMIC

Brainstorming: sources of clinical health data

头脑风暴：临床健康数据的来源

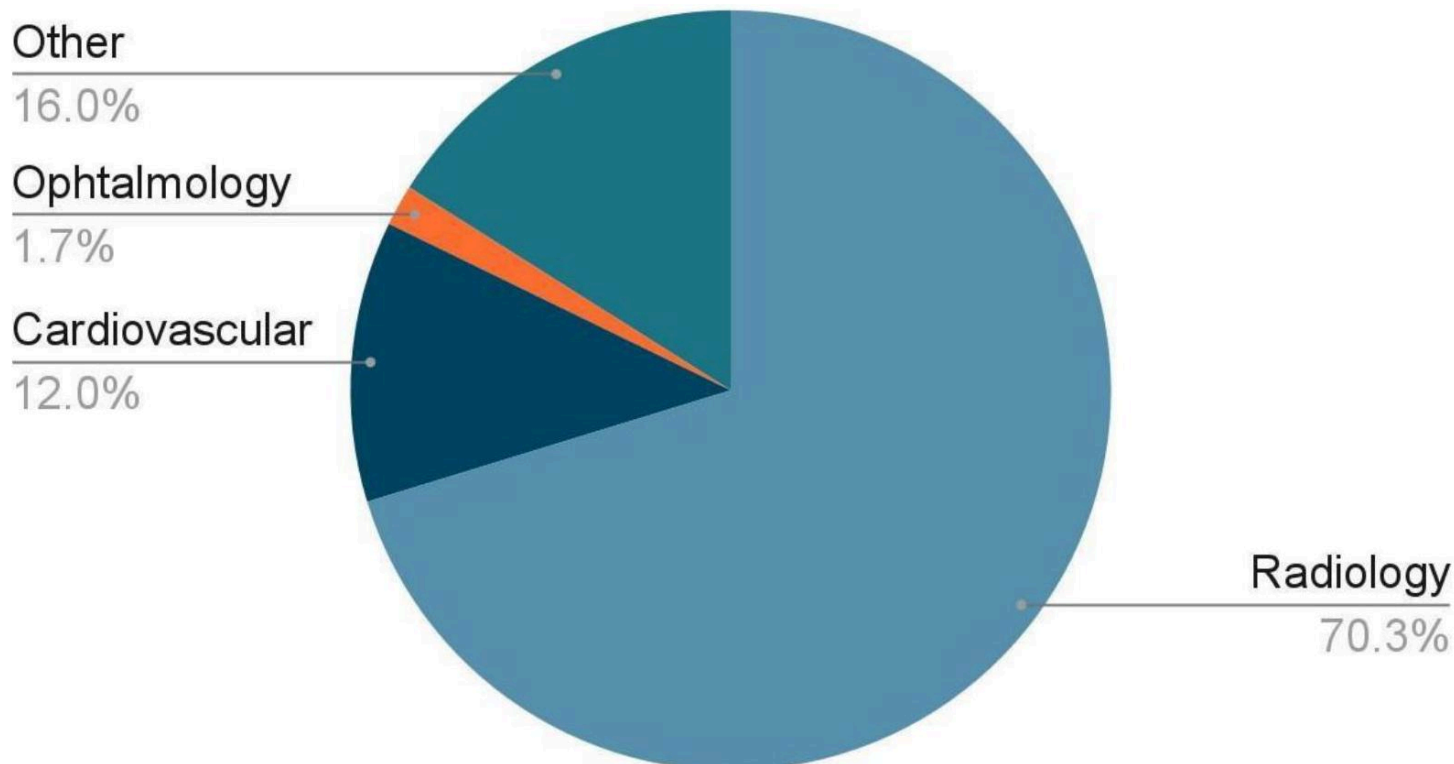
Which areas of Medicine have made the greatest progress in developing/adopting AI?

医学的哪些领域在开发/采用 AI 方面取得了最大的进展？

Which areas of Medicine have made the greatest progress in adopting AI?

医学领域中哪些方面在采用人工智能方面取得了最大进展？

FDA Approvals FDA 批准



Publicly available data sources

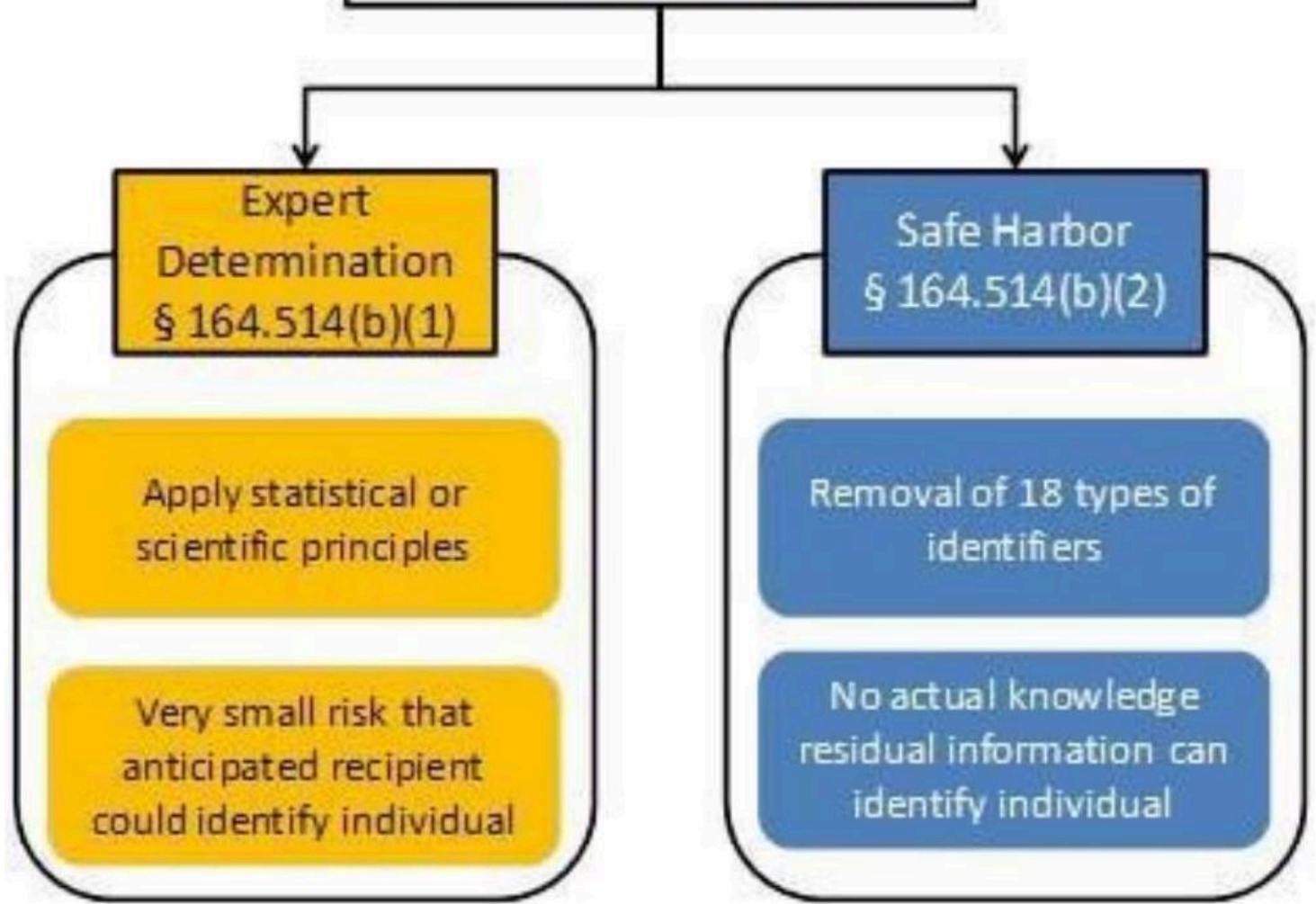
公开可用的数据源

- NOC (nnany) in Austrain 澳大利亚的 NOC (nnany)

In the USA: Health Information Portability and Accountability Act (HIPAA)

在美国：《健康保险可携性与责任法案》（HIPAA）

HIPAA Privacy Rule De-identification Methods



Some datasets available 一些可用的数据集

- MIMIC and its companion datasets in Physionet: MANY!! <https://physionet.org/content/>

Physionet 上的 MIMIC 及其配套数据集：非常多！！ <https://physionet.org/content/>

- US Medicare SynPUF - <https://www.cms.gov/Research-Statistics-Data-and-Systems/Downloadable-Public-Use-Files/SynPUFs>

美国 Medicare SynPUF - <https://www.cms.gov/Research-Statistics-Data-and-Systems/Downloadable-Public-Use-Files/SynPUFs>

- Synthetic data generator - <https://synthetichealth.github.io/synthea/>

合成数据生成器 - <https://synthetichealth.github.io/synthea/>

- Google public data explorer: <https://datacommons.org/>

Google 公共数据浏览器: <https://datacommons.org/>

- Previous Kaggle challenges: <https://www.kaggle.com/datasets?search=health>

往届 Kaggle 挑战赛: <https://www.kaggle.com/datasets?search=health>

- MTSamples (clinical notes): <https://mtsamples.com/>

MTSamples (临床笔记) : <https://mtsamples.com/>

- I2b2/n2c2 clinical NLP data sets: <https://n2c2.dbmi.hms.harvard.edu/>

I2b2/n2c2 临床自然语言处理数据集: <https://n2c2.dbmi.hms.harvard.edu/>

Relevant Australian and Victorian Datasets

澳大利亚及维多利亚州相关数据集

- Victorian Admitted Episodes Dataset (VAED)

维多利亚州住院病例数据集 (VAED)

- Victorian Emergency Minimum Dataset (VEMD)

维多利亚州急诊最低数据集 (VEMD)

- Victorian Integrated Non-Admitted Health (VINAH)

维多利亚州综合非住院医疗数据 (VINAH)

- Centre for Victorian Data Linkage

维多利亚州数据链接中心

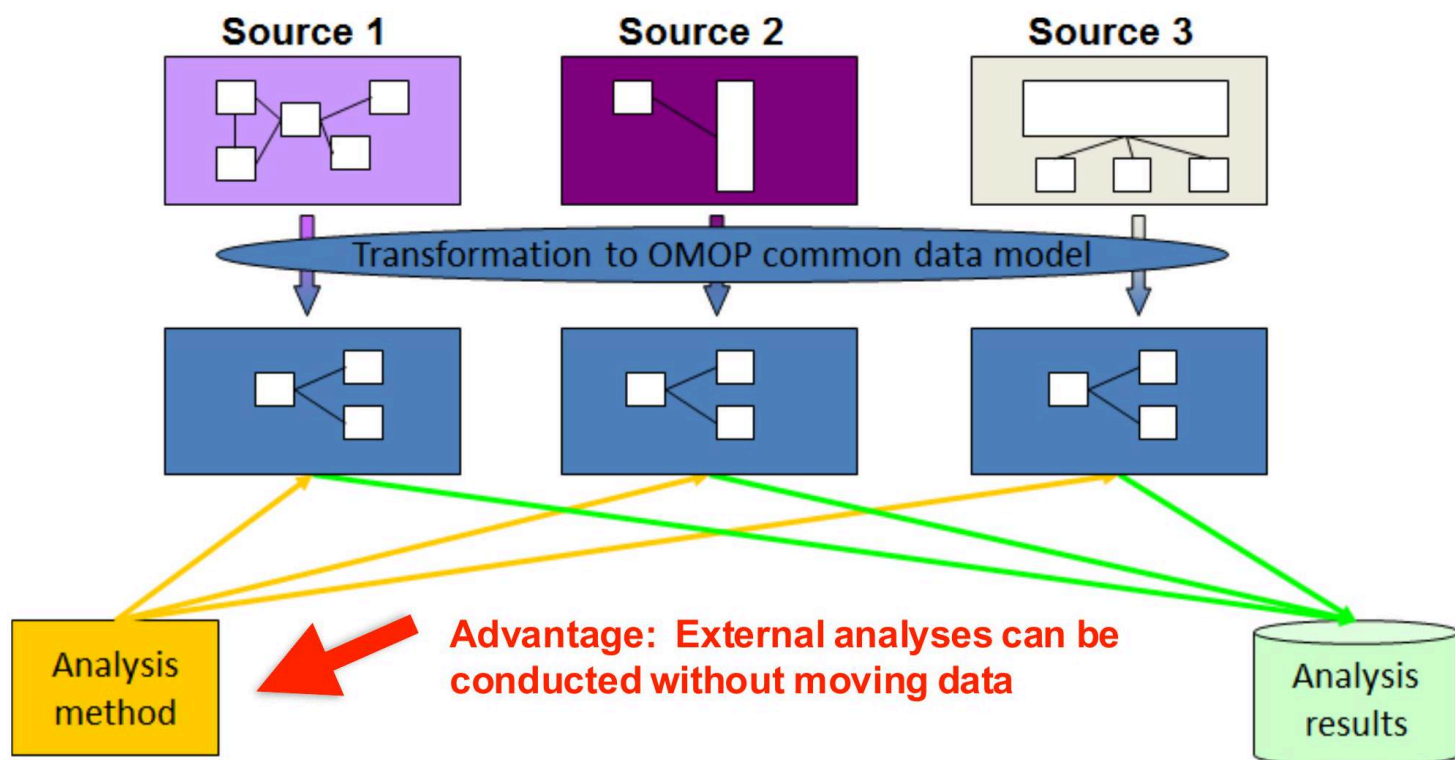
BioGrid

Connecting health information

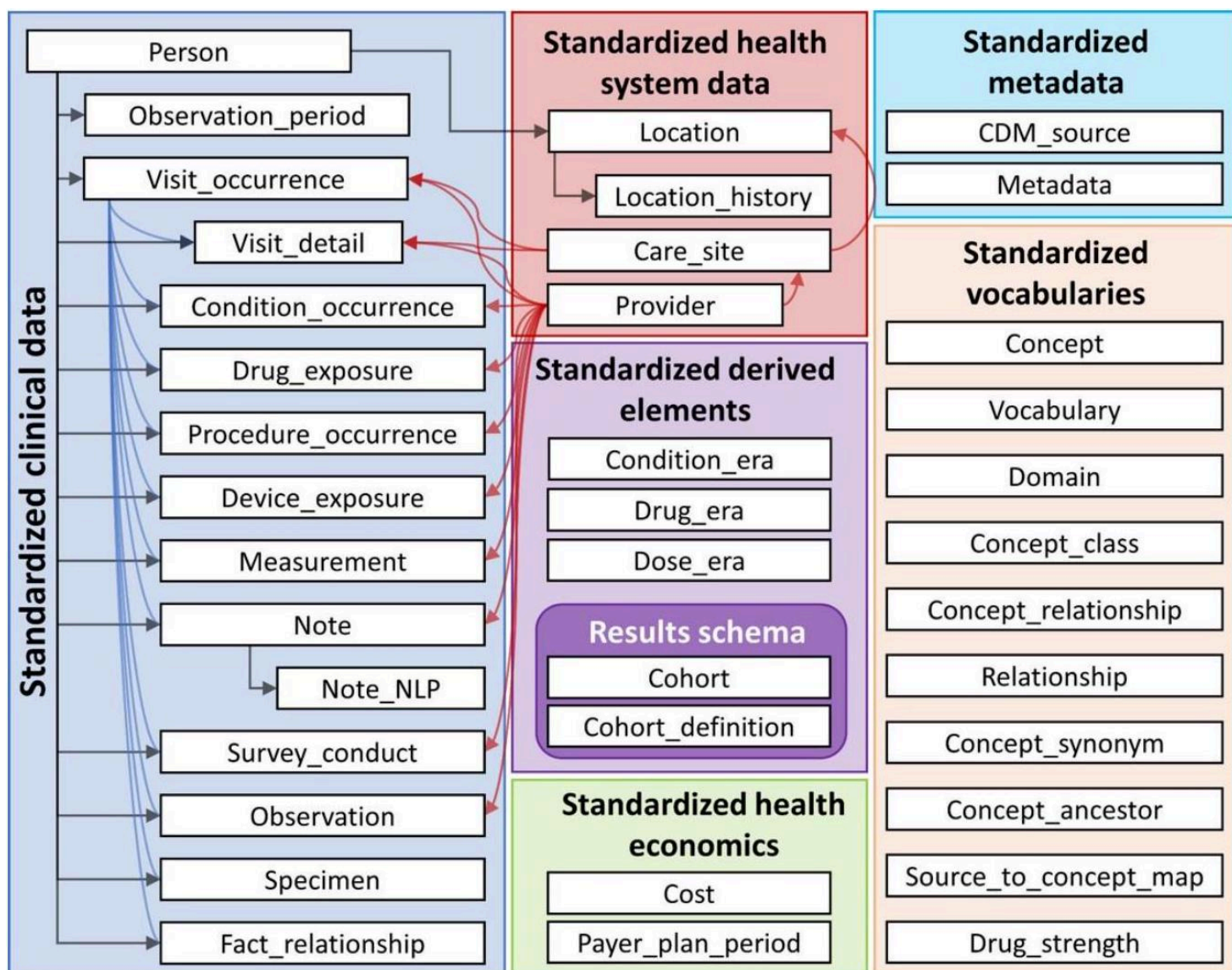
连接健康信息

A word on Common Data Models

关于通用数据模型的说明



OMOP CDM



National COVID Cohort Collaborative

The N3C Data Enclave is a secure platform through which the harmonized clinical data provided by our contributing members is stored. The data itself can only be accessed through a secure cloud portal hosted by NCATS and cannot be downloaded or removed. N3C invites you to begin your journey with the Enclave and join the collaborative efforts of our partners to better understand and address the most pressing COVID-19 clinical questions.

N3C 数据保密库是一个安全平台，用于存储由我们的贡献成员提供的经协调统一的临床数据。数据本身只能通过由 NCATS 托管的安全云门户访问，无法下载或删除。N3C 邀请您开启数据保密库之旅，并加入合作伙伴的协作工作，共同更好地理解 and 解决有关 COVID-19 的最紧迫临床问题。

Help make science go faster and save lives.

帮助加速科学研究，拯救生命。

17.3B

Total Rows

总行数

1,616.3M

Clinical Observations

临床观察

14.8M

1,480 万

Persons

人

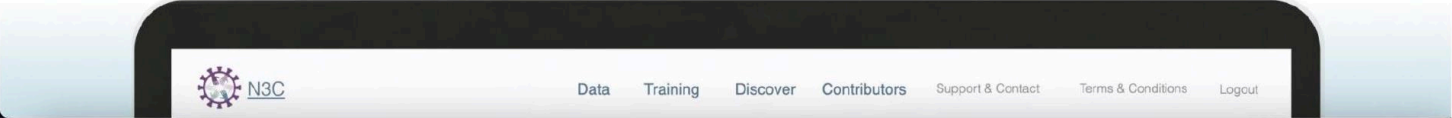
5,815,680

COVID+ Cases

COVID-19 阳性病例

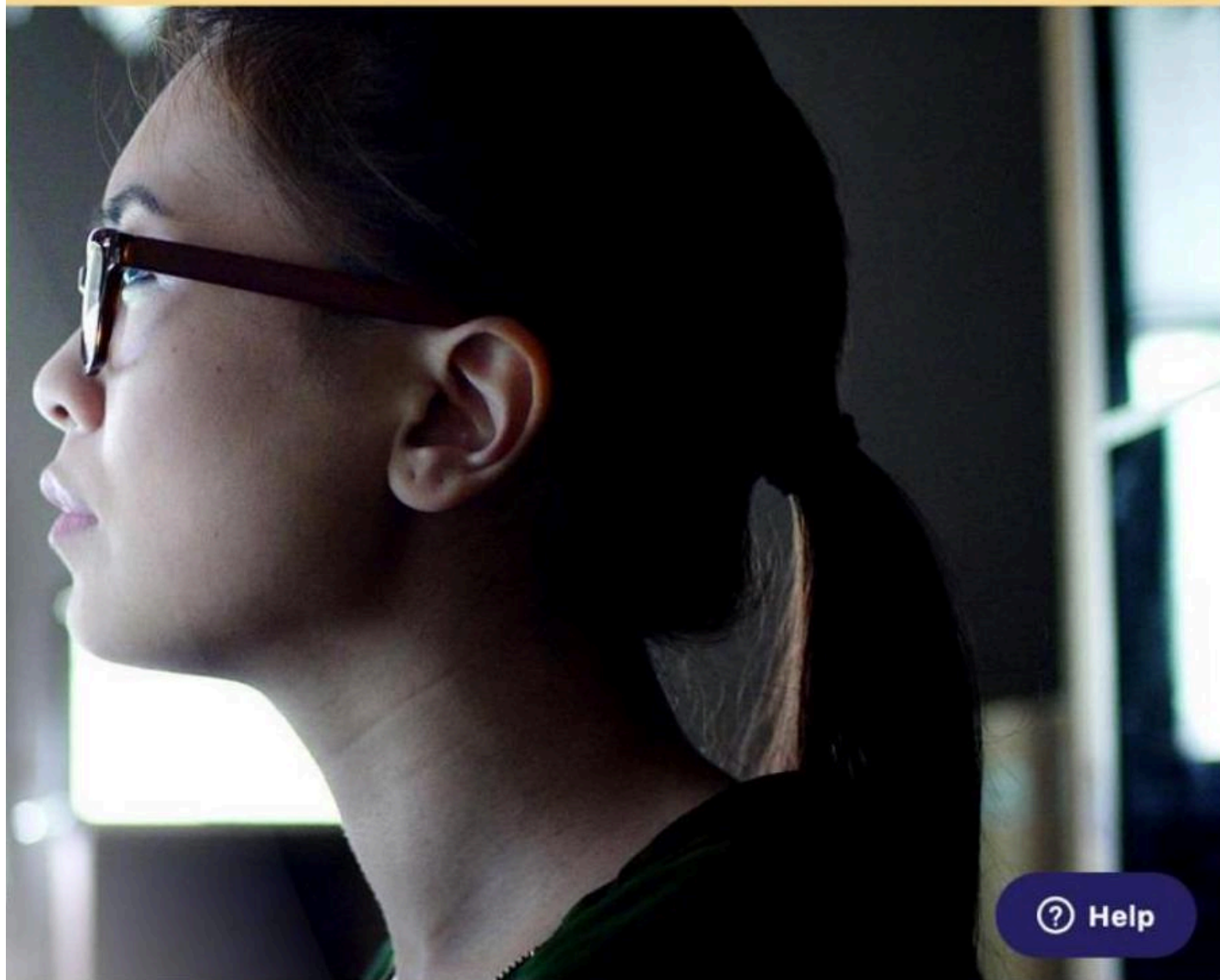
Explore the Full Cohort Dashboard

探索完整队列仪表板





Explore Genomic Data in the Data Browser >>



Project using OMOP 使用 OMOP 的项目



OHDSI

OBSERVATIONAL HEALTH DATA SCIENCES AND INFORMATICS

Welcome to OHDSI Australia

欢迎来到 OHDSI Australia

The Observational Health Data Sciences and Informatics (OHDSI, pronounced “Odyssey”) initiative is an interdisciplinary collaboration involving various stakeholders. Our mission is to unlock the potential of health data by utilizing large-scale analytics. Rest assured, all our solutions are open-source.

观测性健康数据科学与信息学（Observational Health Data Sciences and Informatics，简称 OHDSI，发音为 “Odyssey” ）倡议是一项由各类利益相关者参与的跨学科协作。我们的使命是通过利用大规模分析来释放健康数据的潜力。请放心，我们所有的解决方案都是开源的。

Background 背景

The establishment of OHDSI Australia has been facilitated by close cooperation with the Transformational Data Collaboration (TDC) [https://machaustralia.org/ projects/transformational-data-collaboration/](https://machaustralia.org/projects/transformational-data-collaboration/)

OHDSI Australia 的建立得益于其与 Transformational Data Collaboration (TDC) 的密切合作 [https://machaustralia.org/ projects/transformational-data-collaboration/](https://machaustralia.org/projects/transformational-data-collaboration/)

The TDC operates under the umbrella of the Australian Health Research Alliance (AHRA). Within the AHRA, it falls under the “Data Integration” focus of the “Data Driven Healthcare” activity stream. The TDC is driven by a clear objective:

TDC 在 Australian Health Research Alliance (AHRA) 的框架下运作。在 AHRA 内部，它隶属于 “Data Driven Healthcare” 活动流下的 “Data Integration” 重点领域。TDC 由一个明确的目标推动：

“To harness the unique, open, and collaborative ethos of AHRA in fostering and supporting national data initiatives where an environment promoting openness, inclusivity, and non-competitiveness is paramount.”

“充分发挥 AHRA 独特、开放且协作的精神，促进和支持国家级数据倡议，在这些倡议中，开放、包容和非竞争性的环境至关重要。”

Latest News: 最新消息：

Learn about the EMR to OMOP Project

了解 EMR 到 OMOP 项目

OHDSI Asia Pacific Symposium Highlights

OHDSI 亚太研讨会精彩回顾

Recently, the University of New South Wales had the privilege of hosting the OHDSI Asia Pacific Symposium. We were thrilled to see a diverse turnout of clinicians and data scientists from across the globe. For those who couldn’t attend or wish to revisit the sessions, recordings are now available.

近日，新南威尔士大学有幸主办了 OHDSI 亚太研讨会。来自全球各地的临床医生和数据科学家踊跃参会，令我们倍感振奋。对于未能出席或希望重温会议内容的人员，现已提供会议录播。

OMOP Challenges OMOP 面临的挑战

- Oja et al. transformed Estonian health data (EHR, prescriptions, claims) to OMOP (10% of the population; 150,000 people). 90% of entries successfully mapped (conditions, observations, drugs, measurements)

Oja 等人将爱沙尼亚的健康数据（电子健康记录、处方、理赔数据）转换为 OMOP 格式（覆盖人口的 10%，即 150,000 人）。90% 的条目成功完成映射（疾病、观察结果、药物、测量值）

But there were some challenges!

但也存在一些挑战！

- Which vocabulary to use? SNOMED-CT? CPT4 (Current Procedural Terminology)?

应使用哪种词汇表？SNOMED-CT？CPT4（当前程序术语）？

- Mismatch in granularity (e.g. a text field may say “HPV test”. There are lots of HPV tests (LOINC))

粒度不匹配（例如，文本字段可能写着“HPV 检测”。但 HPV 检测有很多种（LOINC））

- Time of prescription or time of purchase of a drug? Which to transform to OMOP?

药物的处方时间还是购买时间？应将哪一个转换为 OMOP？

Research and Applications

研究与应用

Transforming Estonian health data to the Observational Medical Outcomes Partnership (OMOP) Common Data Model: lessons learned

将爱沙尼亚健康数据转换为观察性医疗结局合作组织（OMOP）通用数据模型：经验教训

Marek Oja, PhD^{1,*}, Sirli Tamm, MSc¹, Kerli Mooses, PhD¹, Maarja Pajusalu, MSc¹, Harry-Anton Talvik, MSc^{1,2}, Anne Ott, MSc¹, Marianna Laht, MD¹, Maria Malk, MSc¹, Marcus Loo, MSc¹, Johannes Holm, MSc¹, Markus Haug, MSc¹, Hendrik Suvalov, MSc¹, Dage Särg, MSc^{1,2}, Jaak Vilo ©, PhD^{1,2}, Sven Laur, PhD¹, Raivo Kolde, PhD¹, Sulev Reisberg ©, PhD^{1,2}

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